Answer **all** the questions in the spaces provided.

1 A ball is thrown horizontally from the top of a building, as shown in Fig. 1.1.

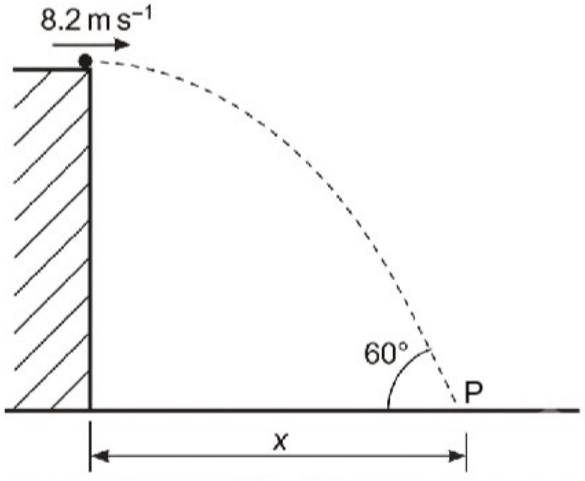


Fig. 1.1

The ball is thrown with a horizontal speed of 8.2 m s^{-1}. The side of the building is vertical. The ball hits the ground at point P at a distance x from the building and at an angle of 60° to the horizontal. Air resistance is negligible.

(a) For the ball at point P,

(i) show that the vertical component of its velocity is 14 m s^{-1},

[2]

(ii) determine the vertical distance it has fallen,

vertical distance = ... m [2]

(iii) determine the horizontal distance x.

distance x = ... m [2]

(b) State and explain the effect on your answer to the horizontal distance x if air resistance is not negligible.

.....

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.....[2]

[Total: 8]